10. Construction Phases, Standard Workflows & Quality Gates

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Primary OWL Classes: `ConstructionPhase`, `Planning`, `Foundation`, `StructuralWork`, `FinishingWork`
Related OWL Classes: `Project`, `Material`, `ConstructionMachine`, `ManagementRole`, `ManualWorker`

10.1 Ontological Representation of Construction Phases

In `stavebna_firma_ontology - kópia.owl`, the lifecycle of every engineering project is structured under the top-level class `ConstructionPhase`. To preserve strict conceptual separation, `ConstructionPhase` is declared formally disjoint with both `Material` and `ConstructionMachine` (`ConstructionPhase owl:disjointWith Material` and `ConstructionPhase owl:disjointWith ConstructionMachine`).

The class `ConstructionPhase` is divided into four sequential subclasses representing the complete execution workflow of a project:

1. **Stage 1: `Planning` (Príprava a projektovanie)**
2. **Stage 2: `Foundation` (Zakladanie stavby)**
3. **Stage 3: `StructuralWork` (Hrubá stavba)**
4. **Stage 4: `FinishingWork` (Dokončovacie práce a odovzdanie)**

10.2 Detailed Breakdown of the Four Construction Phases

Phase 1: Planning (Class: `Planning`)

The `Planning` phase encompasses architectural feasibility, structural statics, BIM modeling, hydrogeological surveys, and building permit acquisition.

- **Key Responsible Roles:** `Architect` (leads commercial planning and client consultations with `Customer`), `Designer` (leads residential engineering for `ResidentialProject`), and `SiteManager` (prepares the Site Mobilization & HSE Plan).
- **Deliverables:** Approved Building Permit, 4D BIM Execution Plan, Bill of Quantities (BoQ) for all `Material` subclasses, and signed procurement schedules with each `Supplier`.
- **Typical Duration & Cost Share:** 15–20% of total timeline | 6–8% of total project budget.

Phase 2: Foundation (Class: `Foundation`)

The `Foundation` phase transfers the building's design loads safely into the load-bearing subsoil.

- **Key Responsible Roles:** `SiteManager` (geodetic layout and concrete pour sign-off) and `ConstructionWorker` crews.
- **Primary Machinery Deployed:** `Excavator` (topsoil stripping, deep excavation pits, utility trenches), `Loader` (gravel sub-base compaction), and `Truck` (soil haulage and transit concrete delivery).
- **Primary Materials Consumed:** Waterproof `Concrete` (C25/30 XC2 to C35/45), B500B reinforcing `Steel` cages, and perimeter XPS `Insulation`.

- **Typical Duration & Cost Share:** 15–20% of total timeline | 12–18% of total project budget.

Phase 3: StructuralWork (Class: `StructuralWork`)

The `StructuralWork` phase erects the primary load-bearing skeleton, floors, roof structure, and external envelope (shell and core).

- **Key Responsible Roles:** `SiteManager`, `ConstructionWorker` (formwork carpenters, ironworkers, masons, timber assemblers), and structural `Architect` / `Designer` for stage inspections.
- **Primary Machinery Deployed:** `Crane` (tower cranes lifting rebar, formwork, steel beams, and CLT panels) and `Truck`.
- **Primary Materials Consumed:**
  - On `EuroveaTower` (`ApartmentBuilding`): High-strength `Concrete` and reinforcing `Steel`.
  - On `FamilyHouseExample` (`FamilyHouse`): Ceramic `Brick` masonry and `Concrete` floor slabs.
  - On `EmpireStateBuilding` & modern `OfficeBuilding`: Structural `Steel` frames and composite slabs.
  - On `WoodenHouse`: Cross-Laminated Timber (`Wood`) panels.
- **Typical Duration & Cost Share:** 30–35% of total timeline | 35–42% of total project budget.

Phase 4: FinishingWork (Class: `FinishingWork`)

The `FinishingWork` phase transforms the structural shell into a fully habitable, energy-efficient, and code-compliant building ready for client handover.

- **Key Responsible Roles:** Specialized `ManualWorker` trades—specifically `Electrician` (wiring, switchboards, lighting, fire alarms on `BuildingProject`) and `Plumber` (water supply, underfloor heating, drainage on `ResidentialProject`)—supervised by the `SiteManager` and inspected by the `Customer`.
- **Primary Materials Consumed:** Triple-glazed `Glass` windows and curtain walls, thermal and acoustic `Insulation`, interior plasters, flooring, and MEP fixtures.
- **Typical Duration & Cost Share:** 25–35% of total timeline | 35–45% of total project budget.

10.3 Summary Matrix: ConstructionPhase vs. Roles, Machines & Materials

Ontology Phase (`ConstructionPhase`)	Lead Ontology Personnel	Dominant `ConstructionMachine`	Dominant `Material` Subclasses	Mandatory Quality Gate Sign-Off
1. `Planning`	`Architect`, `Designer`, `Customer`	Survey vehicles (pre-mobilization)	Specification of all 6 material classes	Gate QG-1: Building Permit & BIM Approval
2. `Foundation`	`SiteManager`, `ConstructionWorker`	`Excavator`, `Loader`, `Truck`	`Concrete`, `Steel` (rebar), `Insulation` (XPS)	Gate QG-2: Geodetic & 28-Day Concrete Cube Test

3. `StructuralWork`	`SiteManager`, `ConstructionWorker`	`Crane`, `Truck`, `Loader`	`Concrete`, `Brick`, `Steel`, `Wood`	Gate QG-3: Structural Topping-Out & Statics Audit
4. `FinishingWork`	`Electrician`, `Plumber` (`ManualWorker`), `SiteManager`	`Truck`, `Loader` (site landscaping)	`Glass`, `Insulation`, MEP piping & cables	Gate QG-4: Pressure/ Electrical Revision & Occupancy Permit